1. Forecasting Using Time Series Analysis

R Implementation Report — May 2026
Dataset 1 (Task 1): Johnson & Johnson Quarterly EPS (1960–1980)
Dataset 2 (Tasks 2–3): AirPassengers — Monthly Airline Passengers (1949–1960)

2. Task 1 — Exponential Smoothing (Johnson & Johnson)

2.1. 1.1 Data Selection & Time Series Construction

Dataset: Johnson & Johnson quarterly earnings per share (EPS), 1960 Q1 to 1980 Q4. The series contains 84 quarterly observations with a clear upward trend and no seasonal fluctuations, making it suitable for exponential smoothing without seasonal components.
The series was split into a training set (76 obs, 1960–1978 Q4) and a control sample (8 obs, 1979–1980 Q4, i.e., the last two years).

2.2. 1.2 Brown's Exponential Smoothing (1st & 2nd Order)

Brown's one-parameter methods use a single smoothing constant α for both level and trend components.

2.2.1. Brown 1st-order (double ES — linear trend)

Implementation: two exponentially smoothed series S't and S''t are computed recursively:

$S't = \alpha \cdot Yt + (1-\alpha) \cdot S't-1$

$S''t = \alpha \cdot S't + (1-\alpha) \cdot S''t-1$

Level: $at = 2S't - S''t$. Trend: $bt = \alpha/(1-\alpha) \cdot (S't - S''t)$. Forecast: $Ft+m = at + bt \cdot m$.
Optimal α found by grid search over [0.01, 0.99] minimising MAD on the 8-period control sample.

2.2.2. Brown 2nd-order (triple ES — quadratic trend)

Three smoothed series S', S'', S''' are computed. The forecast includes a quadratic term: $Ft+m = at + bt \cdot m + 0.5 \cdot ct \cdot m^2$, where $ct = \alpha^2/(1-\alpha)^2 \cdot (S't - 2S''t + S'''t)$.
Optimal α (Brown 1st): 0.17 (MAD = 2.50)
Optimal α (Brown 2nd): 0.10 (MAD = 2.32)

2.3. 1.3 Holt Model (1st-Order Polynomial)

Holt's linear trend model uses two parameters: α (level) and β (trend):

$Lt = \alpha \cdot Yt + (1-\alpha)(Lt-1 + Tt-1)$

$Tt = \beta(Lt - Lt-1) + (1-\beta)Tt-1$

Optimal parameters found by grid search over α, β ∈ [0.01, 0.99] minimising SSE on the control sample.
Result: α = 0.13, β = 0.99 (SSE = 36.55). The high β indicates the trend adapts rapidly to recent changes.

2.4. 1.4 Model Quality Evaluation (MAPE)

MAPE = mean(|actual − forecast| / actual) × 100%, computed on the control sample.

Model	α	β	MAPE
Brown 1st-order	0.17	—	17.85%
Brown 2nd-order	0.10	—	17.07%
Holt	0.13	0.99	14.78%

2.5.1.5 Forecast — 6 Periods Ahead

Holt model (lowest MAPE = 14.78%) was refitted on the full series to forecast 6 quarters ahead (1981 Q1 – 1982 Q2).

Period	EPS Forecast ($)
1981 Q1	15.16
1981 Q2	15.16
1981 Q3	15.17
1981 Q4	15.17
1982 Q1	15.17
1982 Q2	15.17

3. Task 2 — SARIMA Modelling (AirPassengers)

3.1.2.1 Time Series Plot, ACF & PACF, Stationarity Assessment

Dataset: Monthly international airline passengers (thousands), Jan 1949 – Dec 1960, n = 144, frequency = 12.

The time series plot reveals a pronounced upward trend with increasing seasonal amplitude. The ACF exhibits slow decay (indicating non-stationarity in mean) and prominent peaks at lags 12, 24, 36 (indicating annual seasonality). The PACF shows a strong spike at lag 1.

Conclusion: The series is non-stationary due to trend in mean and growing seasonal variance.

3.2.2.2 Differencing Order & ADF Test

A log transformation was applied to stabilise variance. ADF tests were performed on:

- Original log-series: $p \leq 0.01 \rightarrow$ rejected H_0 of unit root (but this is unreliable given obvious trend)
- 1st difference of log: $p \leq 0.01 \rightarrow$ stationary
- 2nd difference of log: $p \leq 0.01 \rightarrow$ stationary (over-differenced)

Correlograms of the 1st-differenced log series show no remaining trend pattern. The 2nd difference introduces excessive negative autocorrelation. Therefore d = 1 is selected.

3.3. 2.3 ARIMA(p,d,q) Order Identification

Analysis of ACF/PACF of the 1st-differenced log series:

- ACF: significant spike at lag 1, then cuts off → suggests MA(1), q = 1
- PACF: significant spike at lag 1 → suggests AR(1), p = 1

Identified model: ARIMA(1,1,1) with log transformation.

3.4. 2.4 ARIMA(1,1,1) Model Construction

Fitted via Arima() from the forecast package with lambda = 0 (log transform).
Coefficients: AR(1) = −0.578 (s.e. 0.130), MA(1) = 0.848 (s.e. 0.087)
AIC = −242.63, AICc = −242.45, BIC = −233.74

3.5. 2.5 Residual Diagnostics of ARIMA(1,1,1)

Ljung-Box test on ARIMA residuals: χ^2 = 245.82, p ≈ 0.000. Significant autocorrelation remains at lags 3, 4, 8, 12, 16, 20, 24 — indicating unmodelled seasonal structure. The ARIMA(1,1,1) alone is insufficient.

3.6. 2.6 Seasonal Component Detection

The ACF of the 1st-differenced series retains strong spikes at lag 12. The seasonal cycle s = 12 (monthly data, annual pattern). A seasonal difference (lag-12) was applied, confirmed to remove the seasonal autocorrelation.

3.7. 2.7 SARIMA(p,d,q)(P,D,Q)$_s$ Construction

Based on the seasonal ACF/PACF of the differenced series:

- Seasonal ACF spike at lag 12 → SMA(1), Q = 1
- Seasonal differencing needed → D = 1

Built SARIMA(1,1,1)(0,1,1)$_{12}$ with log transformation.

3.8. 2.8 SARIMA Residual Diagnostics

Ljung-Box (24 lags): χ^2 = 25.04, p = 0.245 → residuals are white noise. No further modification needed.

3.9. 2.9 Normality of Residuals

Shapiro-Wilk test: W = 0.9836, p = 0.082 → fail to reject normality at α = 0.05. Residuals are approximately normally distributed. Histogram and Q-Q plot confirm this.

3.10. 2.10 Ljung-Box White Noise Verification

Ljung-Box Q-test across lags 6, 12, 18, 24, 30, 36: all p-values > 0.22. The residuals series is confirmed as white noise.

Lag	p-value
6	0.227
12	0.519
18	0.555
24	0.245
30	0.402
36	0.270

3.11. 2.11 Forecast — 18 Periods Ahead

Using the SARIMA(1,1,1)(0,1,1)$_{12}$ model, a forecast was generated for Jan 1961 – Jun 1962 with 95% confidence intervals. The forecast captures both trend continuation and the seasonal pattern (summer peaks, winter troughs).

3.12. 2.12 Auto-ARIMA Comparison

auto.arima() with exhaustive search (stepwise = FALSE) identified SARIMA(0,1,1)(0,1,1)$_{12}$ as the best model by AIC.

Model	AIC	AICc	BIC	MAPE
Manual SARIMA(1,1,1)(0,1,1)[12]	−481.90	−481.58	−470.40	2.58%
Auto SARIMA(0,1,1)(0,1,1)[12]	−483.40	−483.21	−474.77	2.62%

The auto-ARIMA model has marginally better information criteria. Both models produce nearly identical forecasts, confirming that the seasonal structure (0,1,1)$_{12}$ is the dominant component.

4. Task 3 — Classical Decomposition (AirPassengers)

4.1. 3.1 Indicator Selection

Dataset: Monthly international airline passengers (thousands), Jan 1949 – Dec 1960. Monthly frequency with strong seasonal fluctuations, ideal for classical decomposition.

4.2. 3.2 Time Series Analysis, ACF & PACF

The series shows exponential growth (upward curvature) with seasonal amplitude proportional to the trend level. The ACF decays slowly with pronounced peaks at lags 12, 24, 36, confirming a strong annual seasonal component (s = 12). The PACF has a dominant spike at lag 1.

4.3. 3.3 Seasonal Component: Availability, Length & Type

Seasonal length: s = 12 months.
Seasonal type check via boxplots by month showed that July–August are the peak months and November the trough, consistently across all years. The seasonal amplitude grows proportionally with the level (variance increases over time).
Conclusion: MULTIPLICATIVE seasonality. A multiplicative decomposition (Y = Trend × Seasonal × Random) is appropriate because the seasonal fluctuations scale with the trend level.

4.4. 3.4 Classical Multiplicative Decomposition

Using decompose(type = "multiplicative"), the series was split into:

- Trend: smooth upward curve, accelerating after ~1955. Reflects the post-war boom in commercial aviation.
- Seasonal: consistent multiplicative factors ranging ~0.90–1.20. Peak in July (~1.20), trough in November (~0.90). The pattern is stable across years.
- Random: mean ≈ 1.0, no obvious systematic pattern.

An STL decomposition was also performed for comparison, confirming the same structure.

4.5. 3.5 Error Analysis

The random component (ratio of observed to model) was analysed as the model error.
Shapiro-Wilk test: W = 0.9767, p = 0.023 → rejects normality at α = 0.05.
Kolmogorov-Smirnov test: D = 0.084, p = 0.309 → fails to reject normality.
The KS test suggests approximate normality. The slight deviation detected by Shapiro-Wilk may be due to outliers in the early years.
Error mean = 0.998 (ideal: 1.0 for multiplicative model), error variance = 0.0011.

4.6. 3.6 Model Quality Assessment

Metric	Value
R^2	0.9918
MAPE	2.44%
RMSE	9.88
MAE	6.60

R^2 = 0.9918 indicates the decomposition model explains 99.18% of the variance. MAPE = 2.44% confirms excellent predictive accuracy.

4.7. 3.7 Forecast — 2 Years Ahead with Prediction Interval

Forecasting approach:

- Trend: linear regression on the last trend values (R^2 = 0.989), extrapolated 24 periods forward.
- Seasonal: the 12-month seasonal factor pattern repeated for 1961–1962.
- Multiplicative forecast: Trend × Seasonal.
- 95% prediction interval: derived from the log-error standard deviation, assuming lognormal error distribution.

The forecast captures the continued growth trend with realistic seasonal fluctuations.
The prediction interval widens over time, reflecting increasing uncertainty.

5. Summary

Task	Method	Dataset
Task 1	Exponential Smoothing (Brown 1st, Brown 2nd, Holt)	Johnson & Johnson (quarterly EPS)
Task 2	SARIMA(1,1,1)(0,1,1)[12]	AirPassengers (monthly)
Task 3	Classical Multiplicative Decomposition	AirPassengers (monthly)

Best Model	MAPE
Holt (α=0.13, β=0.99)	14.78%
SARIMA(1,1,1)(0,1,1)[12]	2.58%
Multiplicative (Trend × Seasonal)	2.44%

All R scripts and graphical outputs (21 PNG files) are located in the project directory. The AirPassengers dataset yields substantially better forecast accuracy (MAPE ≈ 2.5%) compared to the Johnson & Johnson EPS series (MAPE ≈ 15%), largely because airline passenger data exhibits a stable, predictable seasonal pattern, while quarterly EPS is inherently more volatile.